

A sharp logarithmic barrier for approximate exponential moments

Research note; independently reviewed

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The endpoint lower bound $K \geq cm \log m$ uses a positive measure whose normalized moments approximate those of $e^{-x} dx$. This note shows that moment accuracy and positivity alone cannot yield an endpoint atom bound of a smaller order than the reciprocal logarithm.

Lemma 1. *For the ordinary Laguerre polynomials*

$$L_j(y) = \sum_{k=0}^j (-1)^k \binom{j}{k} \frac{y^k}{k!},$$

one has $|e^{-y/2} L_j(y)| \leq 1$ for $y \geq 0$.

Proof. Here is a self-contained Gaussian proof of this standard inequality. Give entire functions the norm

$$\|f\|^2 = \frac{1}{\pi} \int_{\mathbb{C}} |f(z)|^2 e^{-|z|^2} dA(z).$$

The monomials $f_j(z) = z^j / \sqrt{j!}$ are orthonormal, by polar integration. For a complex number a , the map

$$(D_a f)(z) = e^{-|a|^2/2 + az} f(z - \bar{a})$$

is an isometry: substituting $u = z - \bar{a}$ in its squared norm cancels all exponential factors except $e^{-|u|^2}$. The coefficient of z^j in $e^{az}(z - \bar{a})^j$ is $\sum_{k=0}^j \binom{j}{k} (-|a|^2)^k / k! = L_j(|a|^2)$. Orthogonality of monomials, or direct integration, therefore gives

$$\langle f_j, D_a f_j \rangle = e^{-|a|^2/2} L_j(|a|^2).$$

Cauchy–Schwarz bounds its modulus by one. Choose $|a|^2 = y$. □

Theorem 1. *For every positive even integer d there is a probability measure ν_d on $[0, \infty)$ with an atom of mass exactly $1/(3d)$ at zero, such that for every integer $k \geq 0$,*

$$0 \leq \int \frac{x^k}{k!} d\nu_d(x) - 1 \leq \frac{2^{-d}}{3d}.$$

The difference is zero for $k < d$. In particular, for every sufficiently small $\delta > 0$ there is a positive probability measure with all normalized-moment errors at most δ and an atom at zero of mass at least $c/\log(1/\delta)$, for an absolute $c > 0$.

Proof. Set

$$h_d(x) = 3e^{-3x} \sum_{j=0}^{d-1} L_j(3x), \quad a = \frac{1}{3d},$$

and define

$$d\nu_d(x) = (e^{-x} - ah_d(x)) dx + a\delta_0(dx).$$

By Lemma 1,

$$|h_d(x)| \leq 3de^{-3x/2}, \quad e^{-x} - ah_d(x) \geq e^{-x} - e^{-3x/2} \geq 0.$$

Thus the measure is positive, independently of the parity of d . The continuous part has no atom, so its atom at zero has mass exactly a .

For $z < 3$, termwise integration of each finite polynomial gives

$$\int_0^\infty e^{zx} 3e^{-3x} L_j(3x) dx = \frac{3}{3-z} \left(\frac{-z}{3-z} \right)^j.$$

Summing the finite geometric series yields

$$H_d(z) := \int_0^\infty e^{zx} h_d(x) dx = 1 - \left(\frac{-z}{3-z} \right)^d.$$

In particular $H_d(0) = 1$, so ν_d has total mass one. For $z < 1$ its moment generating function is

$$\int e^{zx} d\nu_d(x) = \frac{1}{1-z} + a \left(\frac{-z}{3-z} \right)^d.$$

The correction has sign $(-1)^d$. For even d , its Taylor coefficients are zero below degree d , and at degree $k \geq d$ they are

$$a \binom{k-1}{d-1} 3^{-k} \geq 0.$$

Their sum is the value at $z = 1$ of the correction series, namely $a2^{-d}$, so each coefficient is at most that quantity. These coefficients are precisely the normalized-moment errors. For odd d the same construction is positive but all these errors have the opposite sign; choosing even d gives the displayed one-sided version. Finally choose the least even $d \geq \log_2(1/\delta)$. Then $a2^{-d} \leq \delta$, while $a \geq c/\log(1/\delta)$. $\square$

If a finite-support example is desired for any prescribed finite moment range $0 \leq k \leq N$, replace the positive continuous part by its positive Gaussian quadrature rule of degree at least N , keeping the atom at zero. Its density is positive for every $x > 0$, has all moments finite, and has infinite support, so the standard orthogonal polynomial construction applies. The resulting finite positive measure has exactly the same moments through degree N and the same atom.

Taking $\delta = m^{-1/2}$ shows that even control of every normalized moment with error $m^{-1/2}$ permits an endpoint atom of order $1/\log m$. This input is stronger than the available bounds $(2k+1)/\sqrt{m}$ for $k \leq \sqrt{m}$. Thus no argument using only those moment inequalities and positivity can force a power-law improvement in the endpoint atom bound. Additional information from the original monotone comparison tensor is necessary. The measures here are not claimed to arise from such tensors or from fair dice.